

LAB 8

Introduction to Object-Oriented Programming

{OOP}

Encapsulation, Interface

Lab08 Exercises

❖ Lab08.1. Grid Layout

Write a program to create a class “GridManagement” and implements from following “Interface GridLayout”
After all, override all functions and implement its own body statement accordingly.

- Use a 2-dimensional array to manage the (7x7) grid
- The value in each cell must be in between 0 to 9 (Set default values of each cell to zero)
- In GridManagment class, you can have more your own additional function/method if needed

```
interface GridLayout{  
    void insertAtRow(int rowNumber, int[] values);  
    void insertAtColumn(int columnNumber, int values[]);  
    void clearAtRow(int rowNumber);  
    void clearAtColumn(int columnNumber);  
    void updateCell(int rowNumber, int columnNumber, int value);  
    void displayGrid();  
    void clear();  
}
```

Console:

:::: The GRID ::::

```
      0 1 2 3 4 5 6  
      - - - - -  
0 - 0 0 0 0 0 0  
1 - 0 0 0 0 0 0  
2 - 0 0 0 0 0 0  
3 - 0 0 0 0 0 0  
4 - 0 0 0 0 0 0  
5 - 0 0 0 0 0 0  
6 - 0 0 0 0 0 0
```

```
1. Update a single cell  
2. Insert at a row  
3. Insert at a column  
4. Clear at a row  
5. Clear at a column  
6. Clear all  
7. Quit  
Choose an opt: 1
```

Lab08 Exercises

❖ Lab08.2. The Bank

Write a program to create a bank management system and manage accounts, and transaction. You must have two needed classes “**Account**” and “**BankManagment**”

- Withdraw/Deposit (Use **Exceptions** to prevent wrong input):
 - Minimum and maximum balance limits (ensure that the account balance does not fall below or exceed a certain)
 - The input field for the account balance should accept only numeric values Allow only **two decimal points** and prevent users from inputting values such as \$100.1234
 - Deposit range (**0 < amount <= 1000\$**)
- Transferring money must be through ACCOUNT_ID

***** All attributes of class Account must be in private access**

***** Implement Setter and Getter concepts**

Console:

```
:::: The Bank ::::
1.  Account List
2.  Create an account
3.  Deposit to an account
4.  Withdraw from an account
5.  Transfer to another account
6.  Quit
Choose an opt: 1
```

Option: 1

```
:::: Account List ::::
No #1
ACC_ID: 00000000111
ACC_HOLDER: Mesa Mina
ACC_BALANCE: 403.34$

No #2
ACC_ID: 00000000222
ACC_HOLDER: Tola Kompheak
ACC_BALANCE: 803.34$

No #3
ACC_ID: 00000000333
ACC_HOLDER: Tola Mokara
ACC_BALANCE: 603.34$

No #4
ACC_ID: 00000000444
ACC_HOLDER: Tola Seyha
ACC_BALANCE: 203.34$
```

Good luck

 How to submit my Lab works???

 Screenshot your source codes and answers as a single **pdf** file and submit to the **Moodle** with Java file attachment as well.

Note: Don't zip the files for submission